

Assignment

Module 1.1 Overview of Hardware components

1. Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.



Feature	HDD (Hard Disk Drive)	SSD (Solid-State Drive)
Speed	Takes more time to open files and applications.	Opens files and applications quickly.
Durability	More likely to get damaged because it has moving parts.	More durable because it does not have moving parts.
Storage	Provides large storage at a cheaper cost.	Costs more for the same amount of storage.
Power	Requires more power to operate.	Requires less power to operate.
Noise	May produce noise during operation.	Operates without making noise.
Uses	Commonly used for storing videos, pictures, documents, and backups.	Commonly used for operating systems, applications, and games.

3. Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.



CPU (Central Processing Unit)

The CPU is our computer's main processor. It runs all the instructions needed to play games like PUBG or Free Fire. The CPU handles things like character movement, game rules, what enemies do, how objects move, and all the math behind the scenes. A faster CPU means the game responds quicker and runs smoother when things get intense.

RAM (Random Access Memory)

RAM temporarily stores information our game needs while playing. It lets our computer quickly grab data like maps, characters, sounds, and game commands. More RAM means fewer freezes and lets us run other programs at the same time as our game.

GPU (Graphics Processing Unit)

The GPU creates all the pictures and visuals in your game. It draws things like textures, characters, scenery, light, shadows, and special effects. A better GPU makes graphics look nicer and gives us more frames per second, which makes the game feel smoother.

4. Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.



If I wanted to upgrade my PC to run the latest FIFA smoothly, I would upgrade these components:

Graphics Card (GPU): I would upgrade the GPU to get better graphics and smoother gameplay.

Processor (CPU): A better CPU would help the game run faster and handle game calculations more easily.

RAM: More RAM would reduce lagging and help the game run smoothly with other programs.

SSD: An SSD would make the game and PC start and load faster.

Cooling System: Better cooling would prevent the PC from overheating during gaming.

Power Supply (PSU): A suitable PSU would provide enough power for the upgraded components.

Overall, I would mainly upgrade the **GPU, CPU, and RAM** because they have the biggest effect on gaming performance.